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小苏打浓度与煮制时间对意大利面拉面化效果的影响：一项 3 × 3 探索性厨房实验

Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti: A 3 × 3 Exploratory Kitchen Trial

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摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房实验考察小苏打浓度与煮制时间对商品干制意大利面拉面化感官表现的共同影响。实验采用3 × 3处理矩阵，将2、4、6 g/L三种小苏打浓度与10、12、15分钟三种煮制时间组合，共形成九种处理。作者依据弹性、中心熟度、顺滑度、碱香、苦涩感及与拉面风格面条的整体相似度进行结构化描述，并给予10分制综合评分。4 g/L、15分钟处理得分最高（9.5/10）；4 g/L、12分钟得分8.5/10。2 g/L处理的变化总体较弱，而6 g/L处理随时间延长出现苦味、涩感、黏口及肥皂样气味。该结果支持将4 g/L、15分钟作为家庭厨房试制的实用起点，但由于研究为单人评价、未报告独立重复且缺少pH和仪器质构测量，结论应理解为探索性观察。

Abstract: This exploratory kitchen trial examined the combined effects of baking-soda concentration and cooking time on the ramen-style transformation of commercial dried spaghetti. A 3 × 3 matrix combined 2, 4, and 6 g/L baking soda with cooking times of 10, 12, and 15 minutes. The author documented elasticity, center doneness, smoothness, alkaline aroma, bitterness, and overall similarity to ramen-style noodles, then assigned a descriptive 10-point score. The highest score, 9.5/10, was assigned to 4 g/L for 15 minutes; 4 g/L for 12 minutes scored 8.5/10. Treatments at 2 g/L produced limited change, whereas prolonged cooking at 6 g/L produced bitterness, astringency, stickiness, and a soap-like odor. The findings support 4 g/L for 15 minutes as a practical starting condition, but remain exploratory because the trial involved one evaluator, undocumented replication, and no pH or instrumental texture measurements.

Keywords: baking soda; spaghetti; alkaline noodles; ramen substitute; sensory evaluation

1. Background / 背景

Fresh alkaline noodles are not consistently available in every market, whereas dried spaghetti is shelf-stable and widely available. Informal recipes often recommend adding an unspecified amount of baking soda, but concentration and time are rarely compared systematically. The study therefore compared nine concentration-time combinations and documented practical sensory differences.

2. Materials and Methods / 材料与方法

Each treatment used 1 L water. Baking-soda concentrations were 2, 4, and 6 g/L and cooking times were 10, 12, and 15 minutes. One evaluator described springiness, surface smoothness, center firmness, alkaline aroma, bitterness, and ramen-like similarity. Independent replication and randomization were not documented.



Figure 2. Commercial spaghetti and 4 g baking soda.

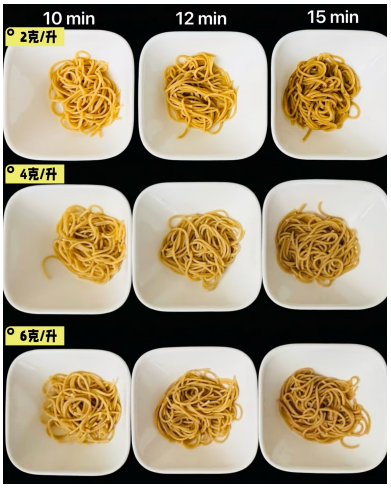


Figure 3. Nine treatments.

3. Results / 结果

g/L	10	12	15
2	5.0	6.0	6.5
4	7.0	8.5	9.5
6	6.0	5.0	3.5

At 2 g/L, scores improved only modestly with time. At 4 g/L, scores increased across time and peaked at 15 minutes. At 6 g/L, scores declined as bitterness, astringency, stickiness, and soap-like odor intensified.



Figure 4. Original score matrix.

4. Practical Procedure / 实用方法

Add 4 g baking soda to 1 L water, bring to a boil, add spaghetti, and cook for 15 minutes. A 12-minute endpoint may suit a firmer preference. Use a larger pot and monitor foaming.

5. Discussion / 讨论

The main pattern was not that more alkali produced a better noodle. A middle concentration had to be balanced with sufficient center softening. Photographic color changes should not be treated as instrumentally calibrated measurements.

6. Limitations / 局限性

Dry pasta mass, replication, randomization, blinding, pH, cooking loss, instrumental texture, and calibrated color were not documented. The results should be interpreted as practical exploratory observations.

7. Conclusion / 结论

Within this trial, 4 g/L for 15 minutes received the highest score. Future work should add standardized mass, replication, independent sensory evaluation, pH, texture, and cooking-loss measurements.

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